《Automotive Configuration》
Chapter 9

Lecturer：Dr. HU Zongjie / Associate Professor

Contact: zongjie-hu@tongji.edu.cn

Chapter 9 Starting System 起动系统

同济大学汽车学院

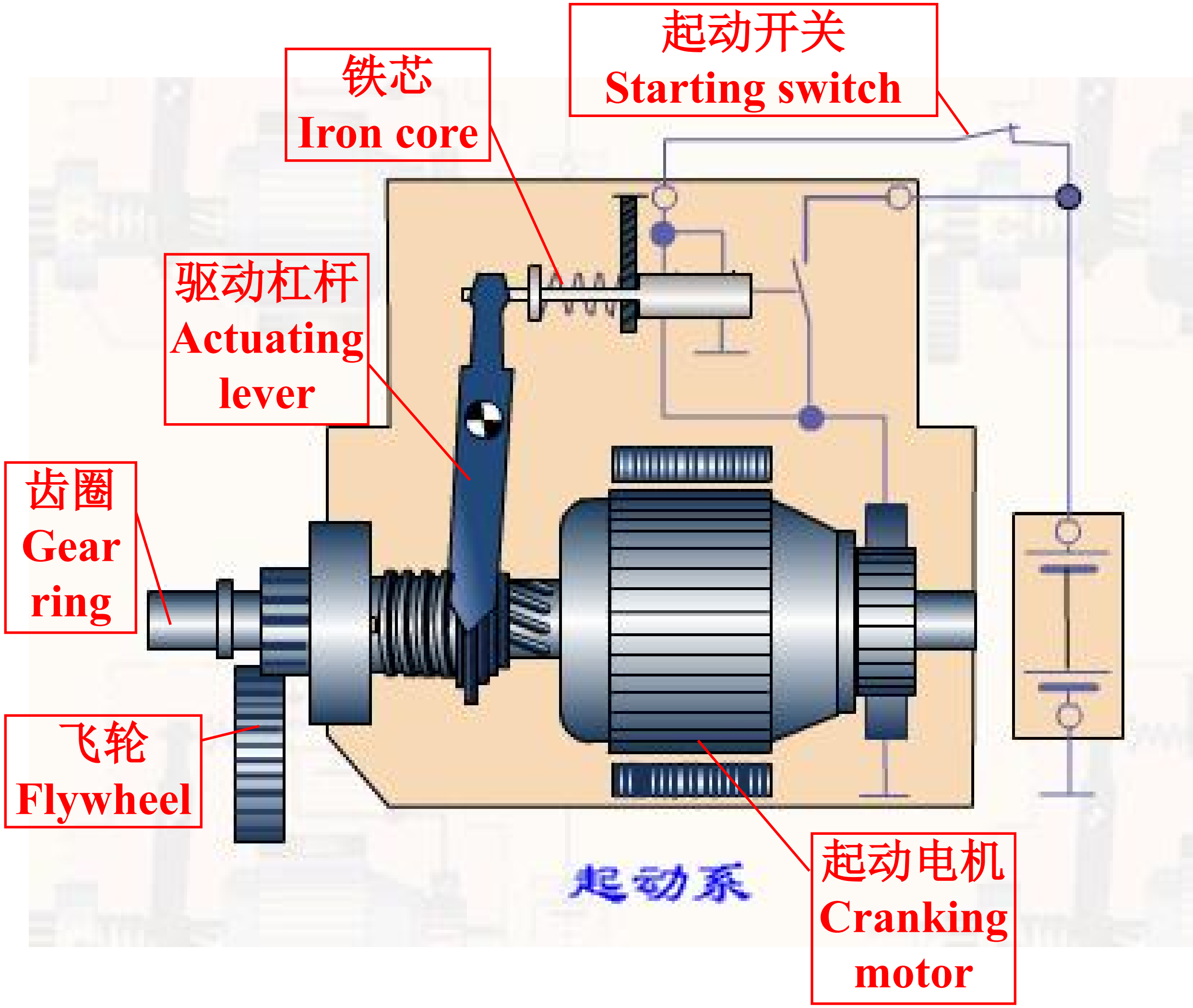
How the one-way clutch works?

What are the differences between normal start system with that for Start/Stop Hybrid?

9.1 Engine Start Overview 发动机起动概述

Definition: Engine start is the process that engine state transits from stop to work. The device used to complete the requirement is called engine starting system.

使发动机从静止状态过渡到工作状态的全过程，叫发动机的起动。完成起动所需要的装置叫起动系 。



1. Starting Conditions起动条件

1) Starting Torque 起动转矩

It is the lowest torque to rotate crankshaft. Starting torque must overcome compression and inner friction resisting moment. The starting resisting moment is related to compression ratio, temperature, oil viscosity, etc.

能够使曲轴旋转的最低转矩称为起动转矩。起动转矩必须克服压缩阻力和内磨擦阻力矩。起动阻力矩与发动机压缩比、温度、机油粘度等有关。

2) Starting Speed 起动转速

It is the lowest speed to rotate crankshaft. At the temperature of 0~20 degree Celsius, the starting speed is 30~40 and 150~300 rpm for gasoline engines and diesel engines, respectively.

能使发动机起动的曲轴最低转速称为起动转速。在0～20℃时，汽油机的起动转速为30～40 r/min，柴油机的起动转速为150～300r/min。

2. Starting Modes 起动方式

1) Manual Starting 人力起动

It is the easiest starting approach that only needs manpower to rotate crankshaft by a lateral pin embedded into the cranking claw seated in the fore-end of crankshaft.

起动最为简单，只须将起动手摇柄端头的横销嵌入发动机曲轴前端的起动爪内，以人力转动曲轴。

2) Motor Starting 电动机起动

Rotate crankshaft by electric motor as the mechanical driving force when the gear on the shaft of motor and flywheel are engaged. The motor also depends on the storage battery as its source.

电动机起动是用电动机作为机械动力，当将电动机轴上的齿轮与发动机飞轮周缘的齿圈啮合时，动力就传到飞轮和曲轴，使之旋转。电动机本身又用蓄电池作为电源。

2. Starting Modes 起动方式

3) Auxiliary Gasoline Engine Starting 辅助汽油机起动

Large volume, complex structure, and being used for diesel engine starting.

体积大，结构复杂，多用于大功率柴油机的起动。

9.2 Start Assist Device 起动辅助装置

Reason: **In the cold winter, oil viscosity and starting resisting moment increases, storage battery working ability and fuel gasification performance decreases.**

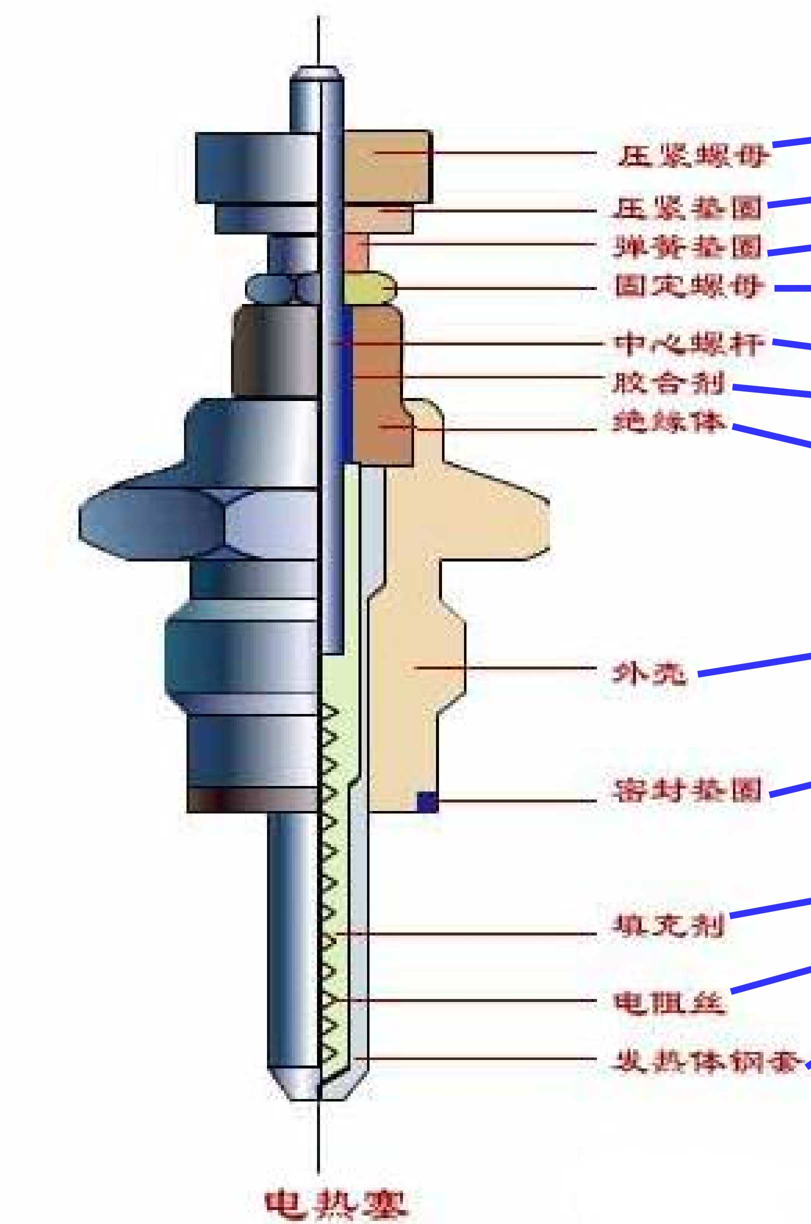
发动机在严寒冬季起动困难，这是由于机油粘度增高，起动阻力矩增大，蓄电池工作能力降低，以及燃油气化性能变坏的缘故。

In order to realize easy cold starting in the winter, intake air, oil and coolant should be preheated. Some auxiliary starting devices which optimize fuel ignition condition and reduce starting torque are equipped by diesel engines.

为使之便于起动，在冬季应设法将进气、润滑油和冷却水预热。车用柴油机为了能在低温下迅速可靠的起动，常采用一些用以改善燃料着火条件和降低起动转矩的起动辅助装置。

Including 这些装置包括: glow plug 电热塞，air preheater 进气预热器，starting fluid spray device 起动液喷射装置，pressure relief device 减压装置

1. Glow Plug 电热塞

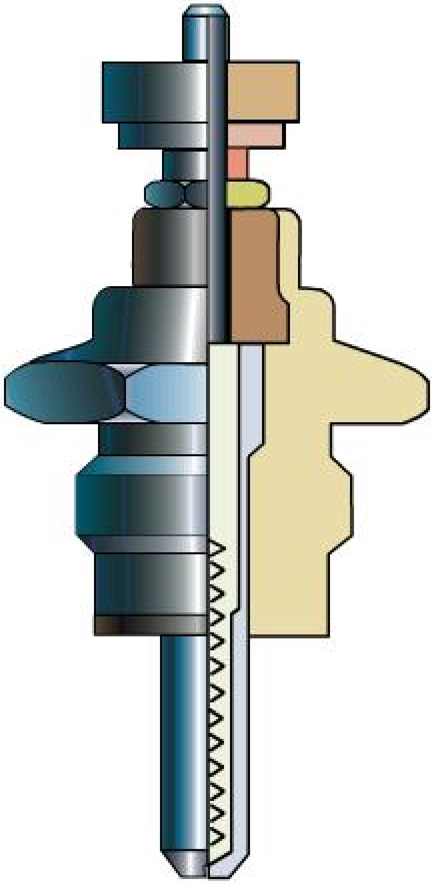


- Gland nut 压紧螺母
- Packing washer 压紧垫圈
- Spring washer 弹簧垫圈
- Fixing nut 固定螺母
- Center nut 中心螺母
- Cement 胶合剂
- Insulator 绝缘体
- Casing 外壳
- Sealing washer 密封垫圈
- Bulking agent 填充剂
- Resistance wire 电阻丝
- Heating body steel sleeve 发热体钢套

1. Glow Plug 电热塞

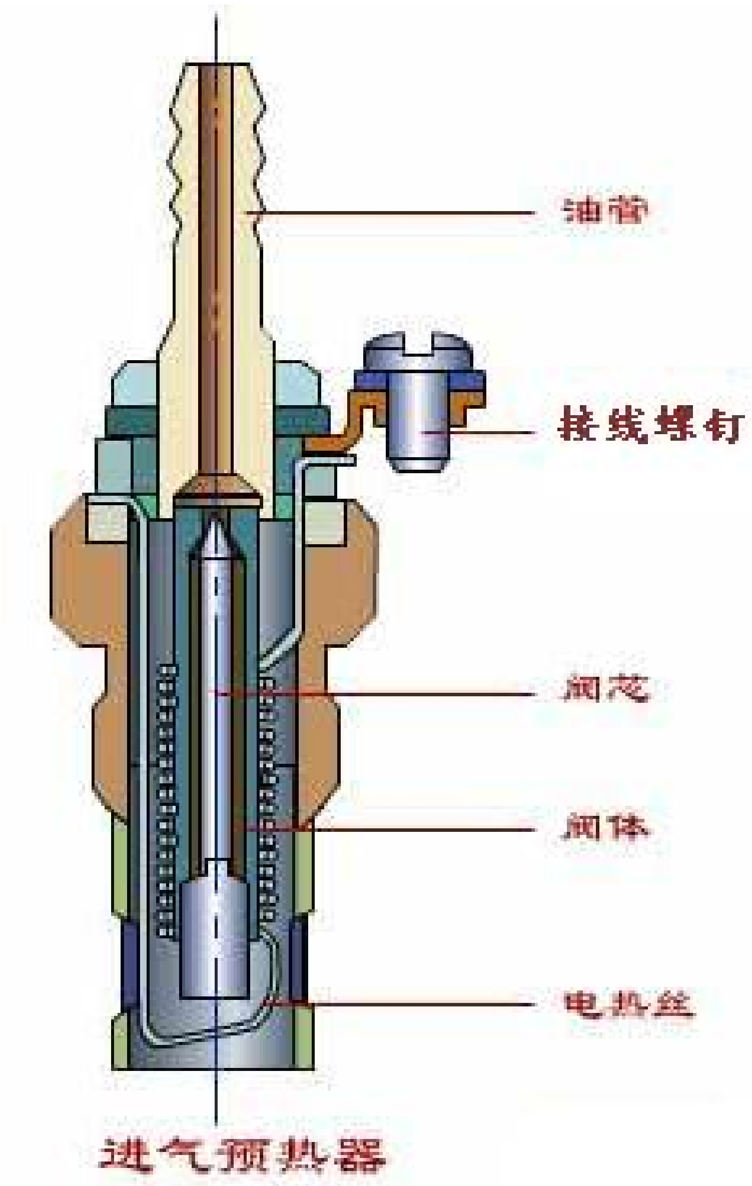
Function: Preheat air in combustion chamber during cold start. In general, glow plug is employed by the divided combustion chamber of diesel engines.

在起动时对燃烧室内的空气进行预热，便于起动。一般在采用涡流室式或预燃室式燃烧室的柴油发动机中装有电热塞。



电热塞

2. Air Preheater 进气预热器



Fuel pipe 油管

Binding screw 接线螺钉

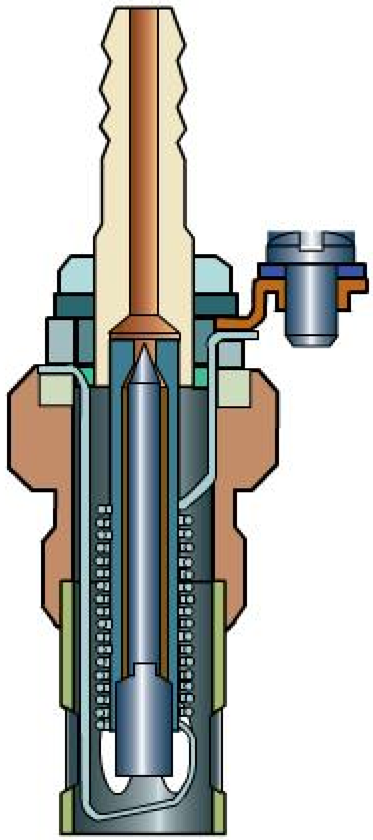
Valve core 阀芯

Valve body 阀体

Electric heating wire 电热丝

2. Air Preheater 进气预热器

Employed by small and medium power diesel engines.
在中、小功率柴油机上常采用进气预热器作为冷起动的辅助装置。
Tapering point of valve core, driven by valve body extension due to electric wire heating, leaves the fuel feed hole. Then fuel enters inner cavity of valve body and blow out after gasification, forming fire after ignition by hot electric wire.
柴油机起动时，接通预热器电路后，电热丝发热，同时加热阀体，阀体受热伸长，带动阀芯移动，使阀芯的锥形端离开进油孔。燃油流进阀体内腔受热气化，从阀体的内腔喷出，并被炽热的电热丝点燃生成火焰喷入进气管，使进气得以预热。


进气预热器

同济大学汽车学院

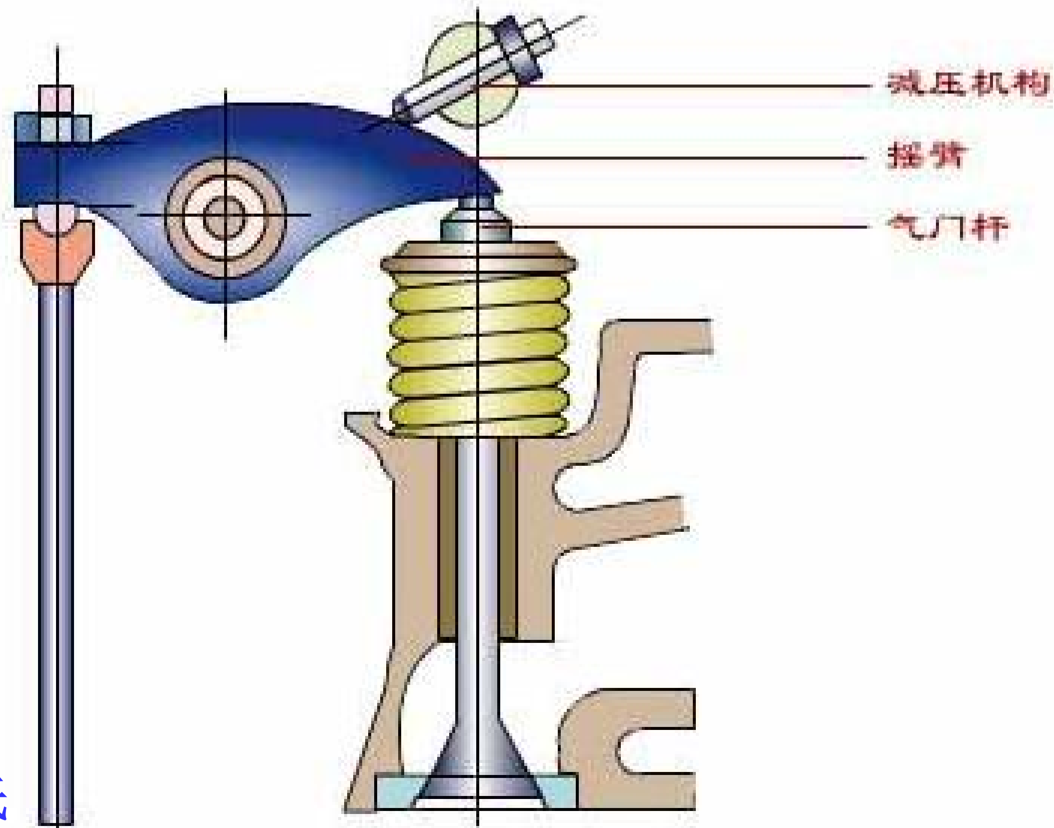
3. Pressure Relief Device 减压装置

To reduce starting torque and increase engine speed for diesel engines.

为了降低起动力矩，提高发动机转速，在某些车用柴油机上采用减压装置。

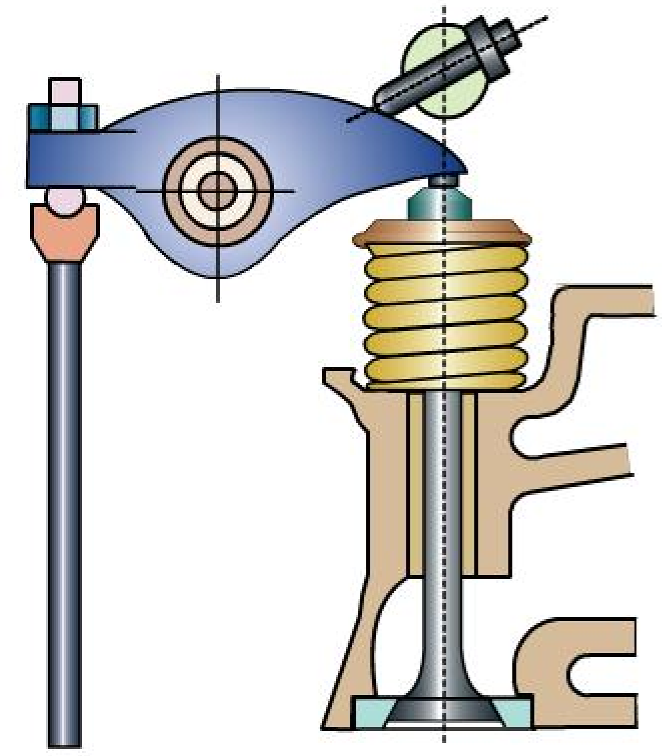
Push valves to open with 1-1.25mm by handle so that starter can rotate crankshaft easily with lower compression resistance.

发动机起动时，首先通过手柄驱使调整螺钉旋转，并略微顶开气门（气门一般下降1-1.25mm），以降低初压缩阻力。这样在柴油机起动前起动机转动曲轴比较容易。



减压装置

3. Pressure Relief Device 减压装置

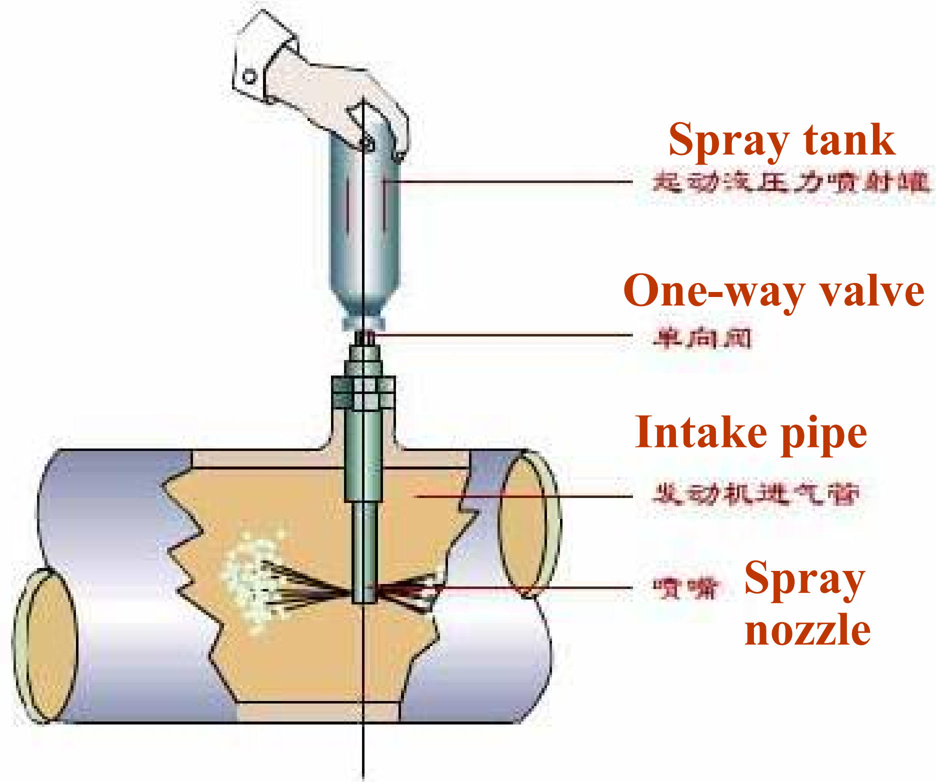


起动减压装置

4. Starting Fluid Spray Device 起动液喷射装置

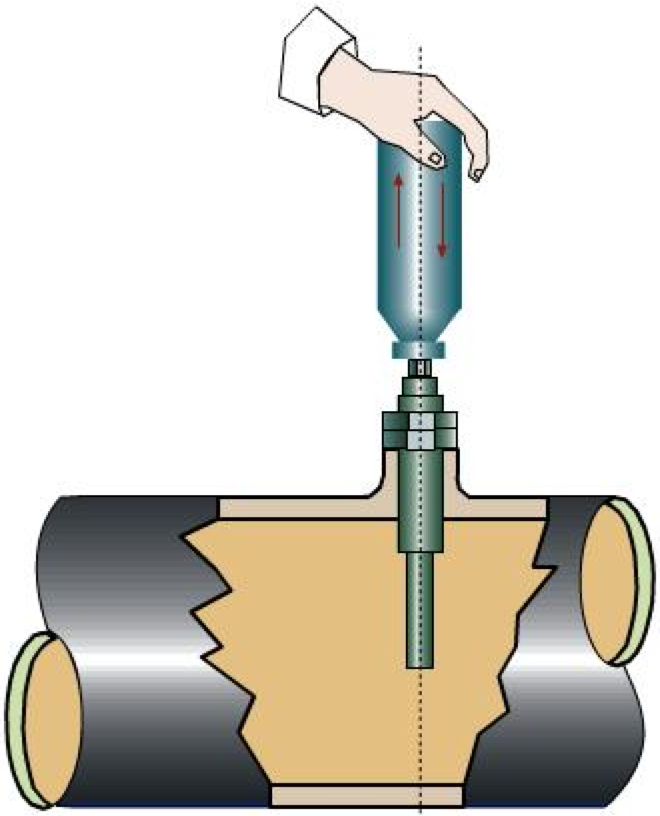
Starting fluid is flammable fuel which can be easily ignited under lower temperature and pressure, to ignite diesel in combustion chamber.

当低温起动柴油机时，将喷射罐倒立，罐口对准喷嘴上端的管口。轻压起动液喷射罐，即打开喷射罐口处的单向阀，则起动液通过单向阀、喷嘴喷入柴油机进气管，并随同进气管内的空气一起被吸入燃烧室。因为起动液是易燃燃料，故可在较低的温度和压力下迅速着火，从而点燃喷入燃烧室的柴油。



起动液喷射装置

4. Starting Fluid Spray Device 起动液喷射装置

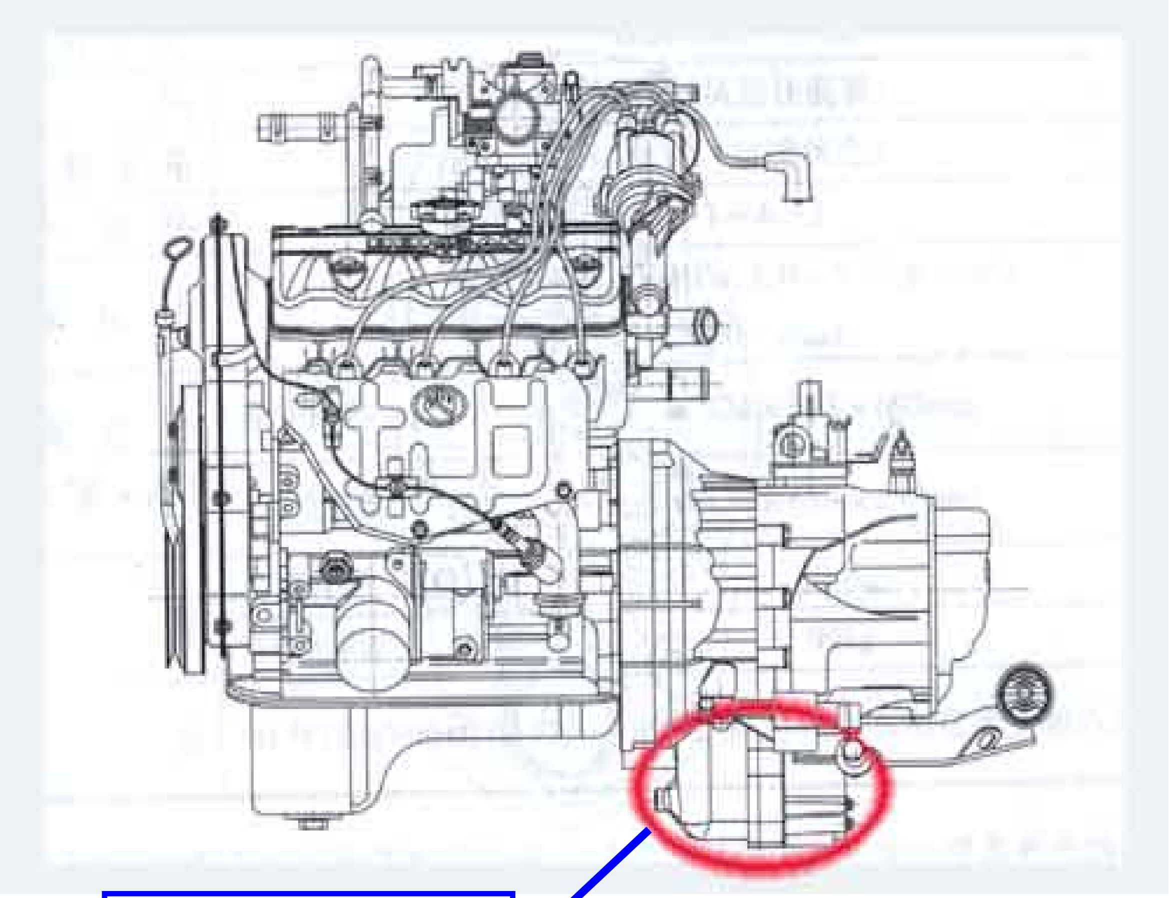


起动液喷射装置

9.3 Starting Motor 起动机

Engine start must have the aid of external force to drive crankshaft to rotate till mixture combustion and do work. Using electric starting motor to crank is almost the only method to assist engine start in modern times.

汽车发动机在起动时，必须用外力带动曲轴旋转，使之完成进气、压缩和点火等过程，直到混合气燃烧做功，发动机才开始工作。电力起动机简称起动机, 用起动机起动发动机几乎是现代汽车的唯一起动方式。

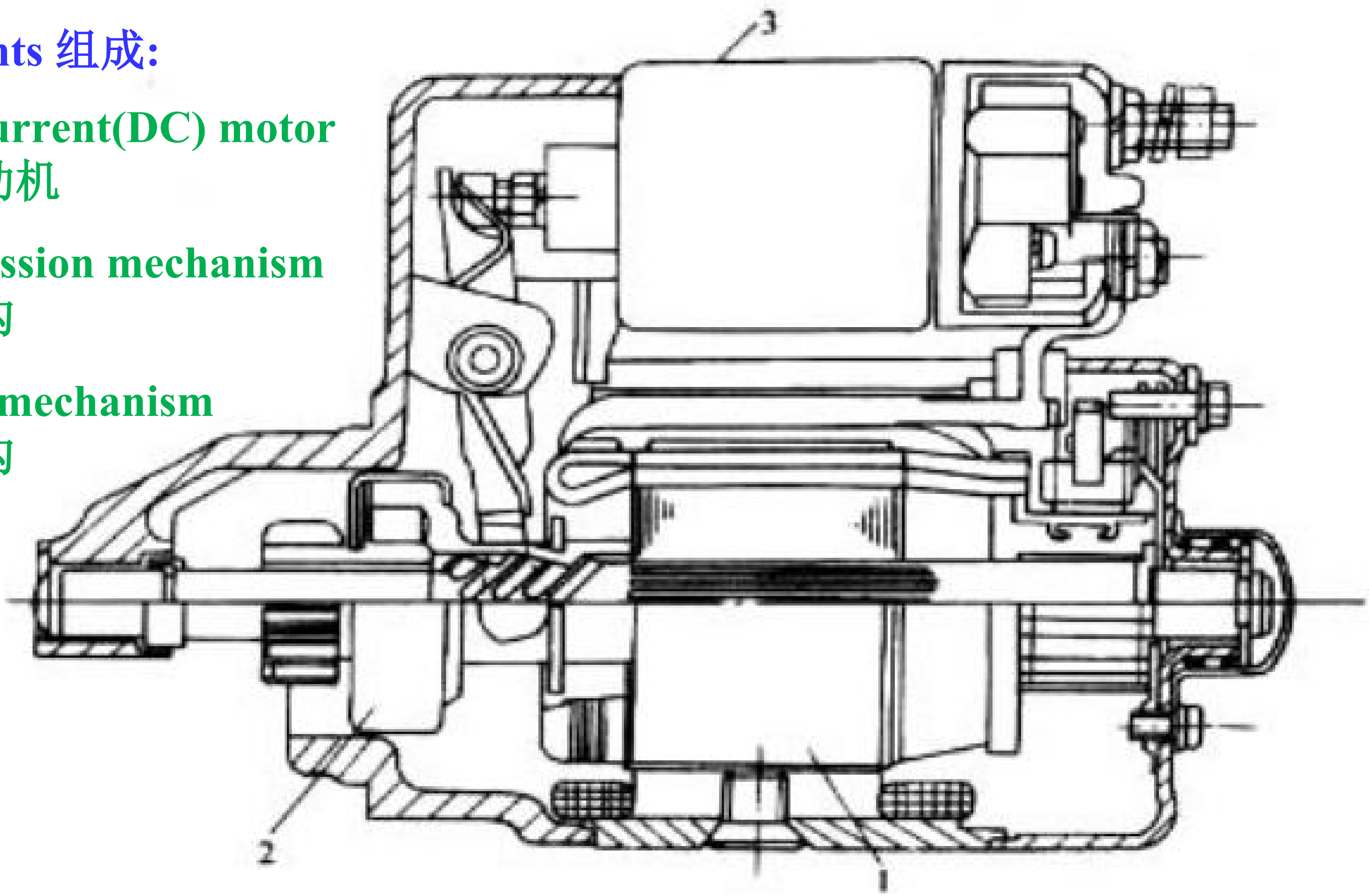


起动机
Starting motor

Components 组成

Components 组成:

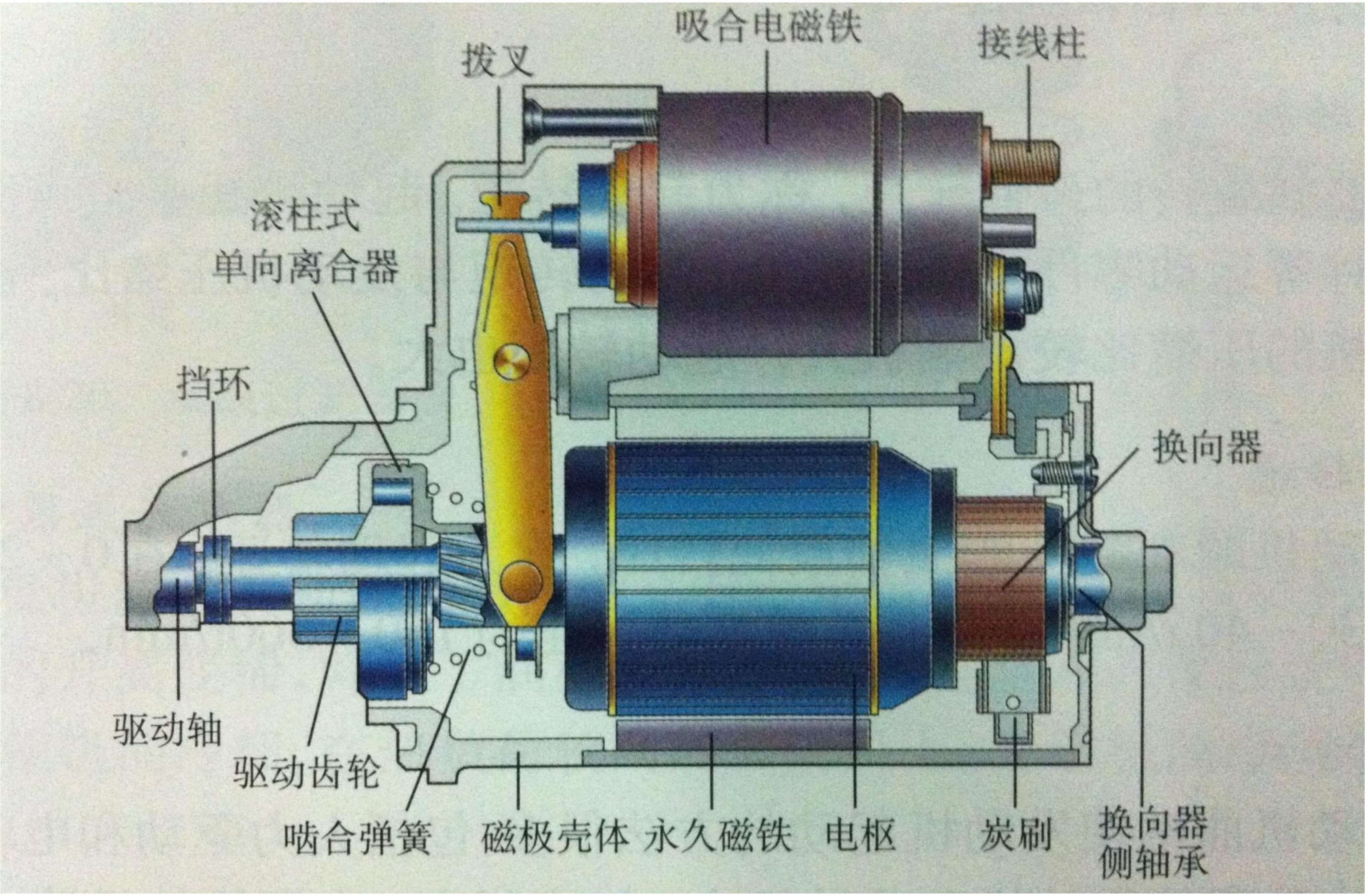
1. Direct current(DC) motor
   直流电动机
2. Transmission mechanism
   传动机构
3. Control mechanism
   控制机构



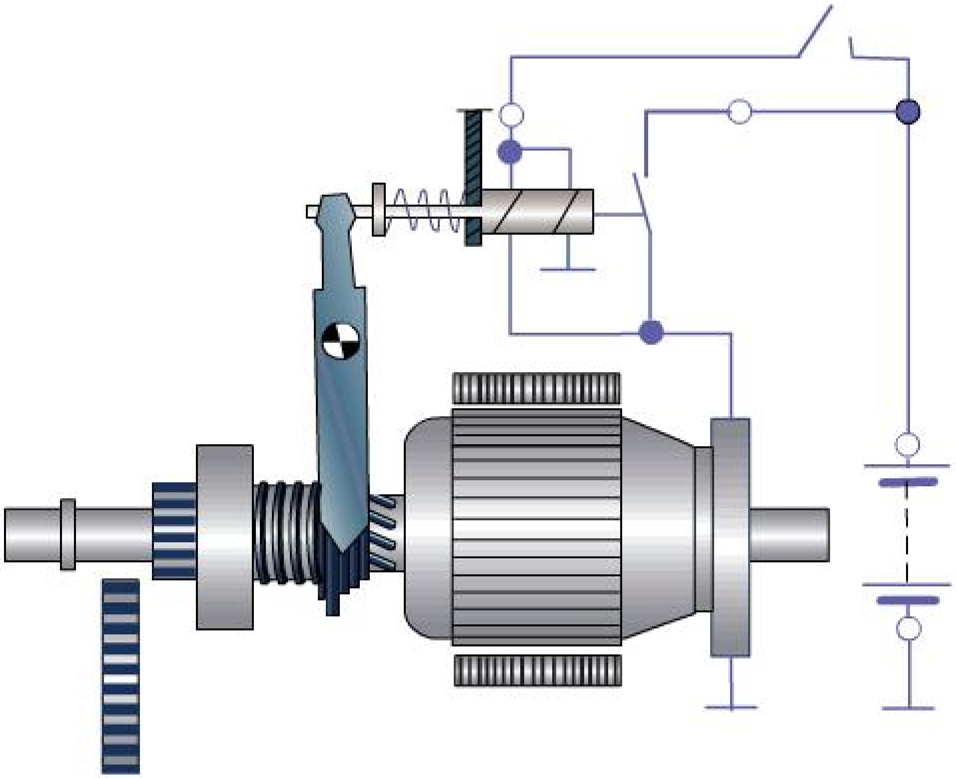
起动机的组成

1—直流电动机 2—传动机构 3—控制机构

Structure结构



Starting Process 起动过程



起动过程

Starting process

1. DC Motor 直流电动机

DC series motor are often used. It has high torque under low speed and the torque will decrease as speed increases, which satisfies the demand of engine.

直流电动机常采用串激直流电动机，其特点是低速时转矩很大，随转速n↑，转矩T↓，这一特征非常适合发动机起动的要求。

For gasoline engine: Power 1.5kW Voltage 12V

For diesel engine: Power 5kW Voltage 24V

2. Transmission Mechanism 传动机构

发动机起动时，将驱动齿轮与电枢轴连成一体，并使驱动齿轮与飞轮齿圈啮合，将起动机产生的转矩传递给发动机曲轴，使发动机起动。

Type:

- Inertial meshing transmission mechanism 惯性啮合式传动机构
- Forced meshing transmission mechanism 强制啮合式传动机构
- Armature mobile transmission mechanism 电枢移动式传动机构

3. Control Mechanism 控制机构

Direct manipulation control mechanism: Drivers manipulate starting switch to clench the transmission gears by starter pedal and lever mechanism. Simple structure, being reliable but inconvenient operation, especially when the driver seats are far away from the starter. Rarely used currently.

直接操纵式控制机构：由驾驶员通过起动踏板和杠杆机构直接操纵起动开关并使传动齿轮副进入啮合。结构简单、使用可靠、但操作不便，且当驾驶员座位距起动机较远时难以布置，目前以很少使用。

Electromagnetic control mechanism: Drivers manipulate relays or starter switch to control electromagnetic switch and gear pairs. It has flexible layout, convenient operation and could be remote controlled for current gasoline and diesel engines. The transmission ratio between starter gear and flywheel is 10-15.

电磁操纵式控制机构：由驾驶员通过起动开关操纵继电器（电磁开关），而由继电器操纵起动机电磁开关和齿轮副或通过起动开关直接操纵起动机电磁开关和齿轮副。 布置灵活、使用方便、适宜于远距离操纵，目前，车用汽油机或柴油机均采用电磁操纵式起动机。起动机齿较与飞轮齿圈传动比为10～15.

9.4 Over-speed Protector 超速保护装置

Also known as one-way clutch, generally including roller type, spring type, plate friction type, etc.

超速保护装置也称为单向离合器，常用的有滚柱式、弹簧式、摩擦片式等多种形式。

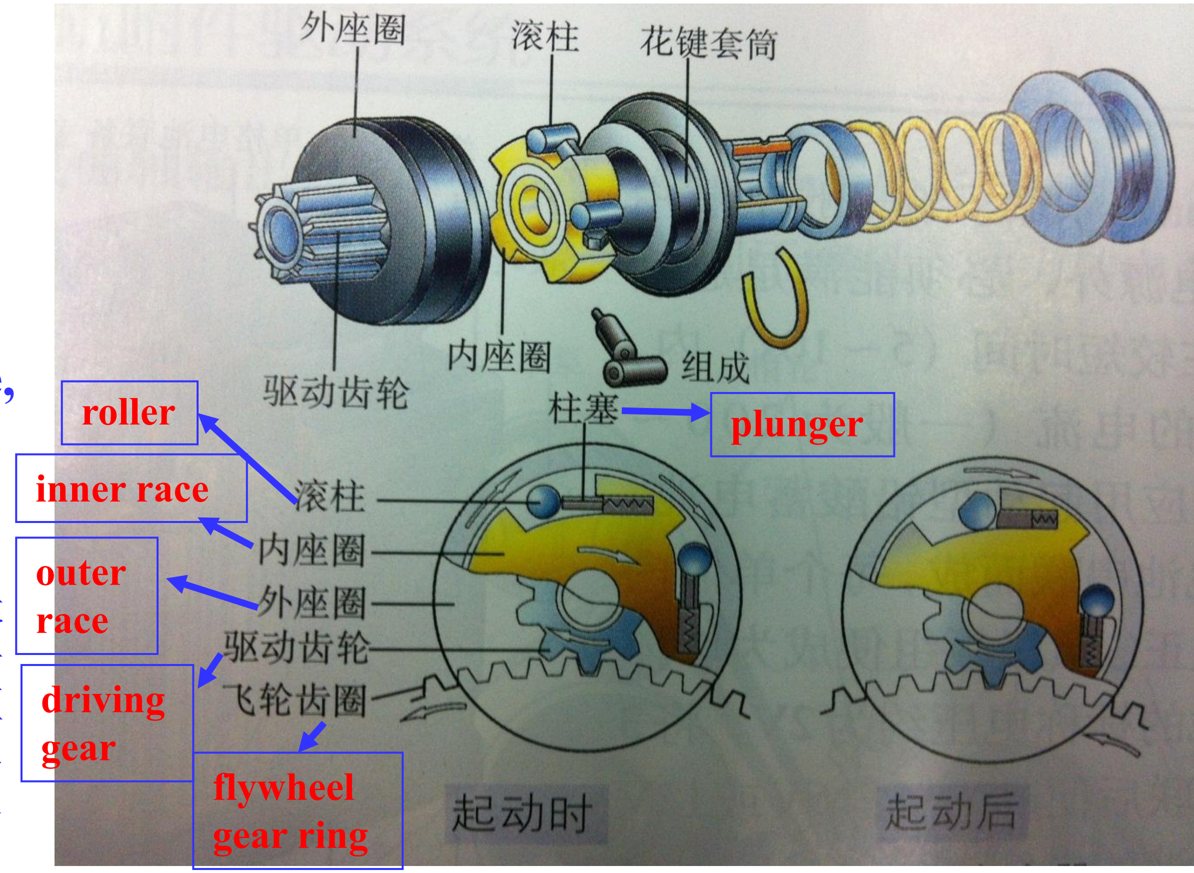
One-way clutch ensures that the driving force of starting motor can transmit from flywheel to crankshaft during start; When the engine begins to operate after start, it can cut off power transmission route with starter, avoiding damage resulted by high speed rotation.

在起动时，它保证起动机的动力能够通过飞轮传递给曲轴；起动完毕，发动机开始工作时，立即切断动力传递路线，使发动机不可能反过来通过飞轮驱动起动机以高速旋转，避免起动机损坏。

Roller Type One-way Clutch 滚柱式离合机构

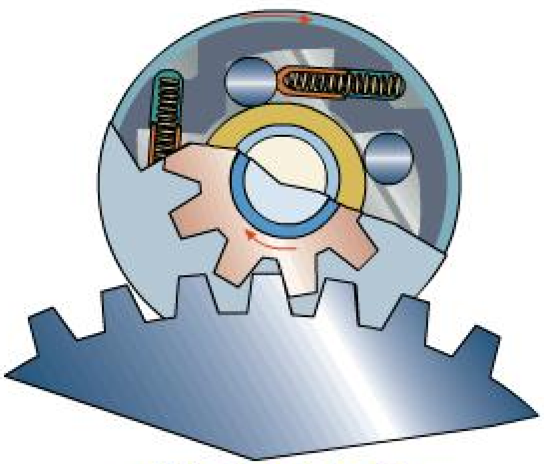
When starting, roller moves to the narrow part of the cuneiform by spring and resistant forces; after starting, the roller moves to the wide part of the cuneiform by resistant force, inner and outer races separate.

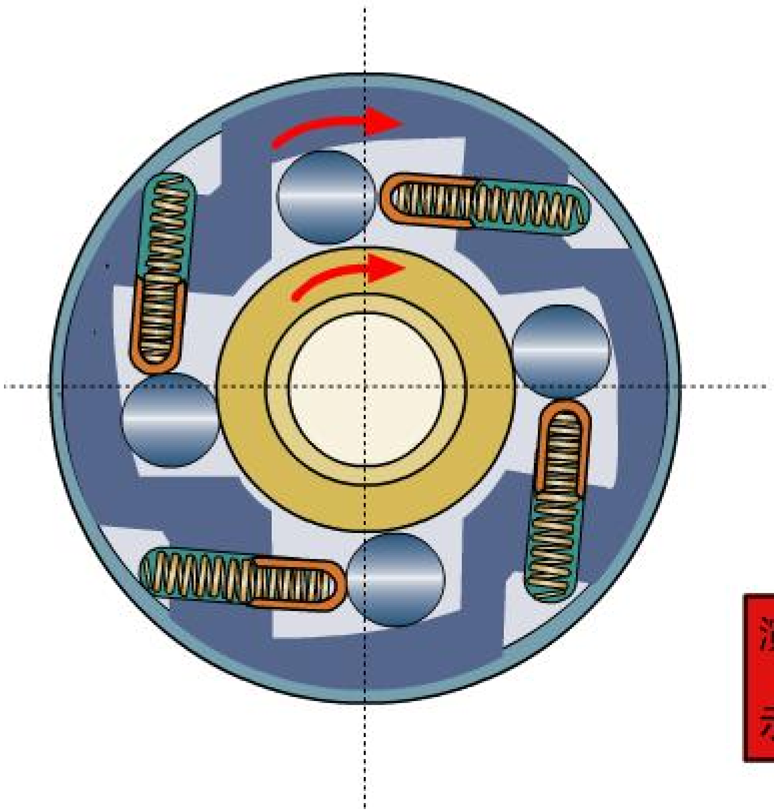
起动时，电枢轴连同内座圈旋转，滚柱受弹簧力和摩擦力作用移动到楔形槽较窄空间，使外座圈与驱动齿轮一起旋转使发动机起动；起动后，飞轮转速超过内座圈与电枢轴，在摩擦力作用下，滚柱克服弹簧作用力向楔形槽较宽空间运动，使内外座圈脱离。



when starting　　after starting

Roller Type One-way Clutch 滚柱式离合机构





演示

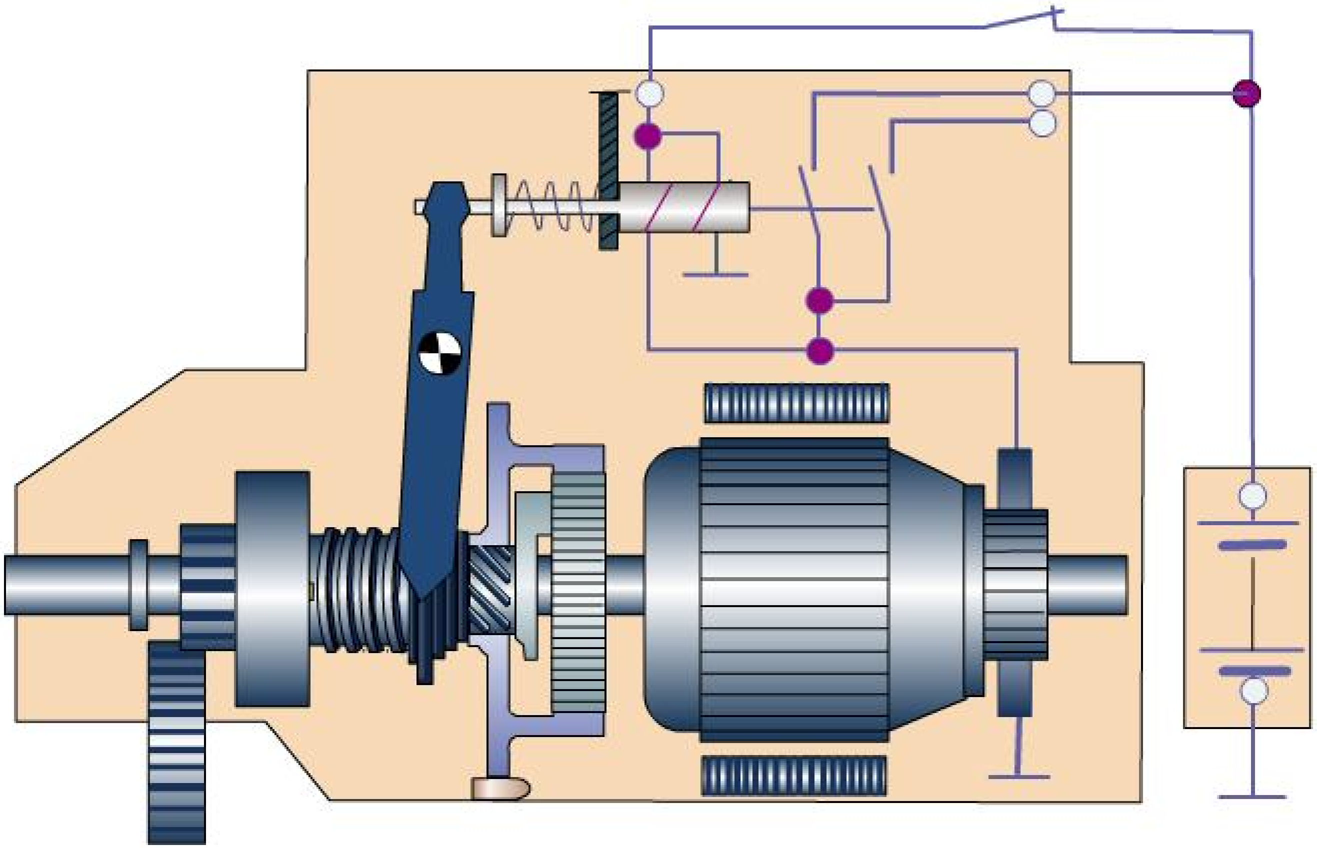
9.5 Decelerating Starter 减速起动机

Decelerating starter has one gear reducer between armature shaft and driving gear.

在起动机的电枢轴与驱动小齿轮之间装有齿轮减速器的起动机称为减速起动机。

If starter power is constant, its volume will reduce with high speed and low torque. So a gear reducer could be used for small, high speed and low torque starter to drive flywheel after transmission. Decelerating starter has smaller volume and weight, and larger driving torque compared to other starters.

当起动机功率一定时，提高电机转速，降低转矩，可以减小其体积。因此，在采用小型、高速、低转矩的起动机时，靠装在电机轴上的齿轮减速器（速比为3-4）将电机转速降低后再驱动飞轮。减速起动机与同功率的起动机相比，它具有体积小、重量轻、驱动转矩大的优点。



减速起动机